Video Compression HW2 – DCT Report

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1. Introduction

I implemented both `2D-DCT` and `two 1D-DCT` approach, and use `PSNR` as evaluation metric.
DCT can transform image into frequency domain and reconstruct image by inverse transform (IDCT).
User can compress image by **discarding high-frequency coefficients** that contribute less to visual perception. This allows for reduction in storage or transmission size with minimal perceptual loss.

2. Computational Complexity Analysis

The computational efficiency of **2D-DCT** and the **two 1D-DCT (accelerated)** approach differs significantly due to the separability property of DCT.

For a square image of size $N \times N$, the **direct 2D-DCT** computes each frequency coefficient $F(u,v)$ using the double summation over all pixels:

$$F(u,v) = \alpha(u)\alpha(v) \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x,y) \cos \frac{(2x+1)u\pi}{2N} \cos \frac{(2y+1)v\pi}{2N}$$

This requires $O(N^2)$ operations for each of the N^2 coefficients, resulting in a total time complexity of:

$$O(N^2 \cdot N^2) = O(N^4)$$

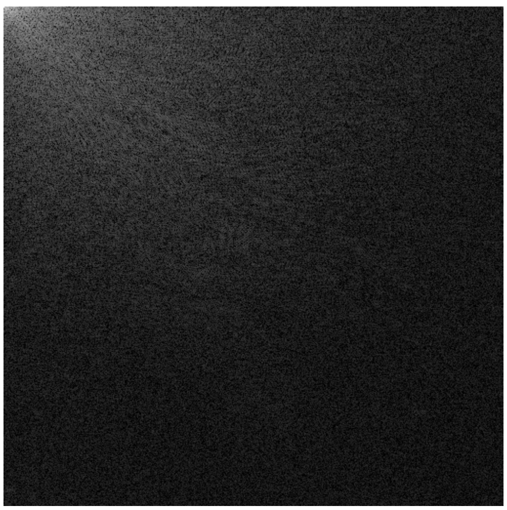
On the other hand, the **two 1D-DCT** approach leverages DCT's separability by first applying 1D-DCT along each row and then along each column. Each 1D-DCT on a row or column of length N requires $O(N^2)$ operations, and with N rows and N columns, the total complexity becomes:

$$O(N^3 + N^3) = O(2N^3) \approx O(N^3)$$

Method	Time Complexity	Notes
2D-DCT (direct)	$O(N^4)$	Simple but computationally expensive for large images
Two 1D-DCT	$O(N^3)$	Faster due to separability; produces identical results

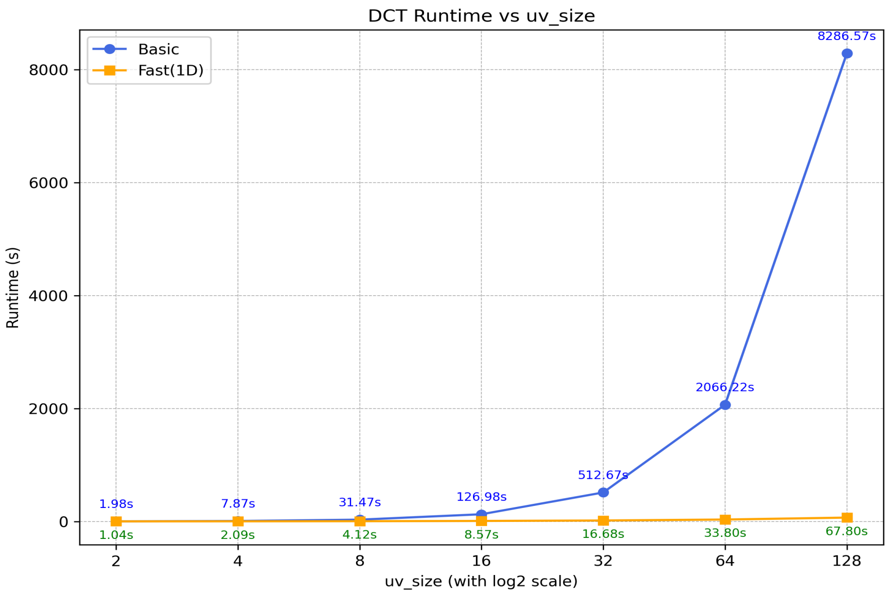
Thus, for large images or real-time applications, the **two 1D-DCT method is significantly more efficient**, while maintaining the same accuracy as the full 2D-DCT.

3. Results

- Grayscale Lena
  
- DCT Coefficients (Log Domain)
  
- Reconstructed Image (Using IDCT)
  

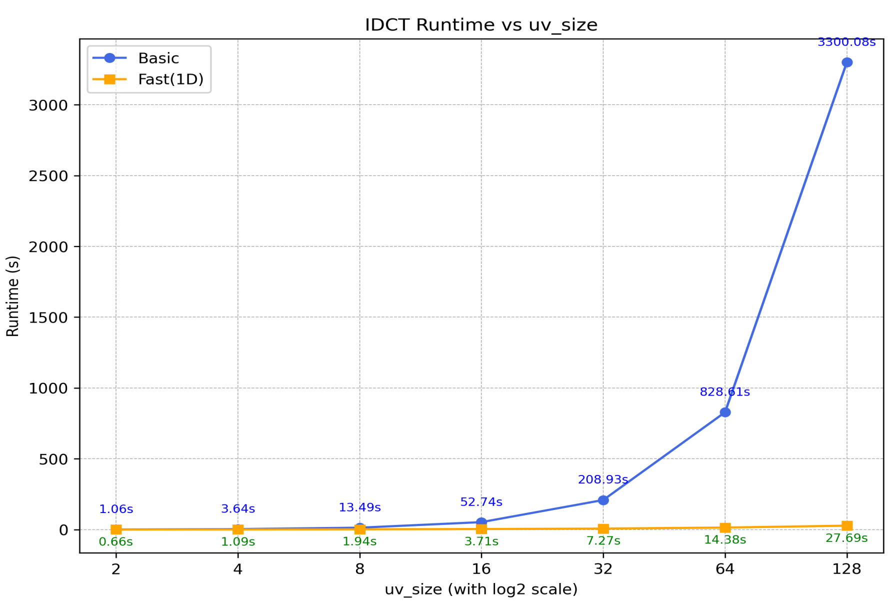
4. Runtime & PSNR Comparision (2D-DCT vs. two 1D-DCT)

DCT Runtime (2D-DCT & two 1D-DCT)



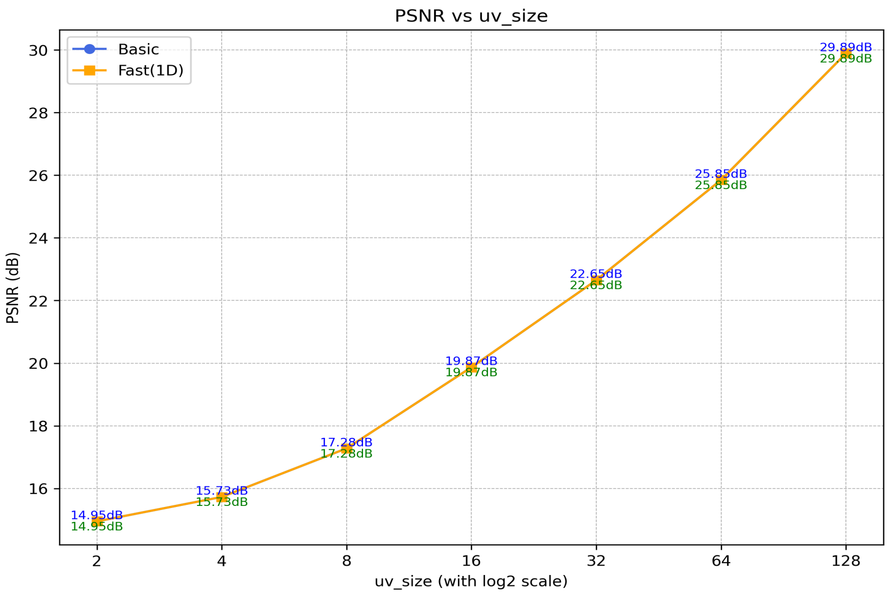
- The **computational complexity** of the 2D-DCT is $O(N^2 \times N^2) = O(N^4)$, since it directly computes the transform over both spatial dimensions simultaneously.
- In contrast, the **two 1D-DCT** approach applies one-dimensional DCT operations sequentially along rows and columns, reducing the complexity to $O(2 \times N^3)$.
- As a result, when the DCT size (`u_size` , `v_size`) doubles,
  - the runtime of **2D-DCT** grows approximately **4×**,
  - while the runtime of **two 1D-DCT** grows only about **2×**.
- From the figure, it is evident that two 1D-DCT achieves a significant speed-up as the block size increases, demonstrating its computational advantage in practice.

IDCT Runtime (2D-DCT & two 1D-DCT)



- A similar trend can be observed for the inverse transform (IDCT).
- The **2D-IDCT** runtime scales quadratically with image size, while the **two 1D-IDCT** scales linearly with each dimension.
- This property makes the two 1D-IDCT more scalable and efficient for larger image sizes.

PSNR (2D-DCT & two 1D-DCT)



- Both methods theoretically produce identical reconstructed images, leading to nearly identical **PSNR** values.
- Minor discrepancies arise from floating-point rounding and computation order differences.
- Overall, the two 1D-DCT achieves the same reconstruction quality while greatly reducing computation time, confirming it as the more efficient implementation.

5. Observation: Effect of DCT Compression

- It can be observed that when using DCT for compression, the transform tends to **discard fine details** while still **preserving the main structural features** of the image.
- As the `uv_size` increases, more frequency components are retained, leading to higher-quality reconstruction.

Reconstructed Images at Different UV Sizes

- `uv_size = 64`

- uv_size = 128

- uv_size = 512